

10.0.0.100 — Flask + Redis через Kustomize: dev и prod.

10.0.0.110 — Prometheus + Grafana + Blackbox через Helm chart, созданный из docker-compose проекта через Kompose.

1. BM 10.0.0.100 — Flask + Redis через Kustomize

Очистка кластера на BM 10.0.0.100

```
root@ubuntu-1:~/lab8# kubectl get all
Command 'kubectl' not found, but can be installed with:
snap install kubectl
root@ubuntu-1:~/lab8#
```

В приложении убираем хардкод redis и выносим имя Redis-сервера в переменную окружения REDIS_HOST.

```
root@ubuntu-1:~/lab8# cat flask_redis/base/flask_redis/app.py
import time
import socket
import os
import redis
from flask import Flask, make_response

DB_HOST = os.getenv('REDIS_HOST', 'redis')
MY_ENV = os.getenv('ENV', 'unknown')

app = Flask(__name__)
cache = redis.Redis(host=DB_HOST, port=6379)

def get_hit_count():
    retries = 5
    while True:
        try:
            return int(cache.get('hits') or 0)
        except redis.exceptions.ConnectionError as exc:
            if retries == 0:
                raise exc
            retries -= 1
            time.sleep(0.5)

def incr_hit_count() -> int:
    return cache.incr('hits')

@app.route('/')
def hello():
    incr_hit_count()
    count = get_hit_count()
    return (
        f'Hello World! I have been seen {count} times. '
        f'My name is: {socket.gethostname()} '
        f'My env: {MY_ENV} '
        f'Redis host: {DB_HOST}\n'
    )

@app.route('/metrics')
def metrics():
    response = make_response(
        f'''# HELP view_count Flask-Redis-App visit counter
# TYPE view_count counter
view_count{{service="Flask-Redis-App", env="{MY_ENV}"}} {get_hit_count()}
'''
    )
    response.mimetype = 'text/plain; charset=utf-8'
    return response
root@ubuntu-1:~/lab8#
```

1.5. Собираем образ внутри Minikube

```
root@ubuntu-1:~/lab8# docker images | grep flask
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
flask:v2                                b47a204f284b          273MB          0B
root@ubuntu-1:~/lab8# minikube image ls | grep flask
docker.io/library/flask:v2
root@ubuntu-1:~/lab8#
```

1.6. Создаём базовые Kubernetes-манифесты

Flask Service

Flask Deployment

Redis Service

Redis Deployment

1.7. Создаём базовую Kustomize-конфигурацию

Проверяем генерацию без применения:

```
root@ubuntu-1:~/lab8# cd ~/lab8/flask_redis

kubectl kustomize base
Command 'kubectl' not found, but can be installed with:
snap install kubectl
root@ubuntu-1:~/lab8/flask_redis# minikube kubectl -- kustomize base
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
    name: redis
spec:
  ports:
    - port: 6379
      targetPort: 6379
  selector:
    app: devops-course-2025
    svc: db
  type: ClusterIP
---
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
    svc: flask-front
    name: service-devops
spec:
  externalIPs:
    - 10.0.0.100
  ports:
    - name: flask-port
      port: 8000
      targetPort: 5000
  selector:
    app: devops-course-2025
    svc: front
  type: LoadBalancer
```

2. Создаём overlay dev

2.1. dev/kustomization.yaml

2.2. Патч сервиса dev

Для dev меняем внешний порт на 54321.

2.3. Патч Deployment dev

Для dev:

2 реплики Flask;

REDIS_HOST=dev-redis;

ENV=DEVELOPMENT.

3. Создаём overlay prod

3.1. prod/kustomization.yaml

3.2. Патч сервиса prod

Для prod меняем внешний порт на 54000.

3.3. Патч Deployment prod

Для prod:

5 реплик Flask;

REDIS_HOST=prod-redis;

ENV=PRODUCTION.

4. Проверяем Kustomize без применения

```
root@ubuntu-1:~/lab8/flask_redis# cd ~/lab8/flask_redis
minikube kubectl -- kustomize dev
minikube kubectl -- kustomize prod
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
    environment: dev
    name: dev-redis
spec:
  ports:
    - port: 6379
      targetPort: 6379
  selector:
    app: devops-course-2025
    environment: dev
    svc: db
  type: ClusterIP
---
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
    environment: dev
    svc: flask-front
    name: dev-service-devops
spec:
  externalIPs:
    - 10.0.0.100
  ports:
    - name: flask-port
      port: 54321
      targetPort: 5000
  selector:
    app: devops-course-2025
    environment: dev
    svc: front
  type: LoadBalancer
---
apiVersion: apps/v1
```

```

root@ubuntu-1:~/lab8/flask_redis# kubectl kustomize prod
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
    environment: prod
    name: prod-redis
spec:
  ports:
    - port: 6379
      targetPort: 6379
  selector:
    app: devops-course-2025
    environment: prod
    svc: db
  type: ClusterIP
---
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
    environment: prod
    svc: flask-front
    name: prod-service-devops
spec:
  externalIPs:
    - 10.0.0.100
  ports:
    - name: flask-port
      port: 54000
      targetPort: 5000
  selector:
    app: devops-course-2025
    environment: prod
    svc: front
  type: LoadBalancer
---
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: devops-course-2025
    environment: prod
    name: prod-flask-app

```

5. Запускаем minikube tunnel на BM 10.0.0.100

```

Sorry, try again.
[sudo] password for user:
root@ubuntu-1:/home/user# sudo kill -f "minikube tunnel" || true

sudo nohup minikube tunnel --bind-address=10.0.0.100 > /tmp/minikube-tunnel.log 2>&1 &

sleep 5

tail -n 20 /tmp/minikube-tunnel.log
[1] 39577
nohup: ignoring input
* Tunnel successfully started

* NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...

root@ubuntu-1:/home/user# █

```

6. Применяем dev и prod окружения

```
kubectl apply -k dev
kubectl apply -k prod

kubectl wait --for=condition=available deployment/dev-flask-app --timeout=180s
kubectl wait --for=condition=available deployment/prod-flask-app --timeout=180s

kubectl get deployments
kubectl get pods -o wide
kubectl get services
service/dev-redis created
service/dev-service-devops created
deployment.apps/dev-flask-app created
deployment.apps/dev-redis created
service/prod-redis created
service/prod-service-devops created
deployment.apps/prod-flask-app created
deployment.apps/prod-redis created
deployment.apps/dev-flask-app condition met
deployment.apps/prod-flask-app condition met
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
dev-flask-app	2/2	2	2	2s
dev-redis	0/1	1	0	2s
prod-flask-app	5/5	5	5	1s
prod-redis	0/1	1	0	1s

```
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
dev-flask-app-5479cf97db-6jjz9 1/1 Running 0 2s 10.244.0.3 minikube <none> <none>
dev-flask-app-5479cf97db-jqtrn 1/1 Running 0 2s 10.244.0.4 minikube <none> <none>
dev-redis-c5d7c4b4-9xz4c 0/1 ContainerCreating 0 2s <none> minikube <none> <none>
prod-flask-app-774f46f78-8mpq7 1/1 Running 0 1s 10.244.0.8 minikube <none> <none>
prod-flask-app-774f46f78-dsh4h 1/1 Running 0 1s 10.244.0.7 minikube <none> <none>
prod-flask-app-774f46f78-g5kkz 1/1 Running 0 1s 10.244.0.11 minikube <none> <none>
prod-flask-app-774f46f78-lfdmk 1/1 Running 0 1s 10.244.0.10 minikube <none> <none>
prod-flask-app-774f46f78-mjd58 1/1 Running 0 1s 10.244.0.6 minikube <none> <none>
prod-redis-688996977d-jhgfz 0/1 ContainerCreating 0 1s <none> minikube <none> <none>
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
dev-redis	ClusterIP	10.98.54.124	<none>	6379/TCP	3s
dev-service-devops	LoadBalancer	10.103.82.50	127.0.0.1, 10.0.0.100	54321:30782/TCP	3s
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	28m
prod-redis	ClusterIP	10.108.50.56	<none>	6379/TCP	2s
prod-service-devops	LoadBalancer	10.99.239.254	127.0.0.1, 10.0.0.100	54000:32494/TCP	2s

root@ubuntu-1:~/lab8/flask_redis#

7. Проверям Flask dev/prod

```
root@ubuntu-1:~/lab8/flask_redis# curl http://10.0.0.100:54321/
curl http://10.0.0.100:54321/

curl http://10.0.0.100:54000/
curl http://10.0.0.100:54000/
Hello World! I have been seen 1 times. My name is: dev-flask-app-5479cf97db-6jjz9 My env: DEVELOPMENT Redis host: dev-redis
Hello World! I have been seen 2 times. My name is: dev-flask-app-5479cf97db-jqtrn My env: DEVELOPMENT Redis host: dev-redis
Hello World! I have been seen 1 times. My name is: prod-flask-app-774f46f78-8mpq7 My env: PRODUCTION Redis host: prod-redis
Hello World! I have been seen 2 times. My name is: prod-flask-app-774f46f78-g5kkz My env: PRODUCTION Redis host: prod-redis
root@ubuntu-1:~/lab8/flask_redis#
```

Проверка метрик:

```
root@ubuntu-1:~/lab8/flask_redis# curl http://10.0.0.100:54321/metrics
curl http://10.0.0.100:54000/metrics
# HELP view_count Flask-Redis-App visit counter
# TYPE view_count counter
view_count{service="Flask-Redis-App", env="DEVELOPMENT"} 2
# HELP view_count Flask-Redis-App visit counter
# TYPE view_count counter
view_count{service="Flask-Redis-App", env="PRODUCTION"} 2
root@ubuntu-1:~/lab8/flask_redis#
```

9. BM 10.0.0.110 — Prometheus + Grafana + Blackbox через Helm/Kompose

9.1. Подготовка и очистка старых ресурсов

```
kubectl config use-context minikube || true

kubectl delete all --all -n default --ignore-not-found=true
kubectl delete configmap --all -n default --ignore-not-found=true
kubectl delete secret --all -n default --ignore-not-found=true
kubectl delete pvc --all -n default --ignore-not-found=true

helm uninstall promgra -n monitoring 2>/dev/null || true
kubectl delete namespace monitoring --ignore-not-found=true

kubectl get all
  > kubectl.sha256: 64 B / 64 B [-----] 100.00% ? p/s 0s
  > kubectl: 49.59 MiB / 49.59 MiB [-----] 100.00% 15.00 MiB p/s 3.5s
Switched to context "minikube".
service "kubernetes" deleted
configmap "kube-root-ca.crt" deleted
No resources found
No resources found
No resources found in default namespace.
root@ubuntu-2:~/lab8#
```

9.2. Устанавливаем Helm

```
helm version

Downloading https://get.helm.sh/helm-v3.21.2-linux-amd64.tar.gz
Verifying checksum... Done.
Preparing to install helm into /usr/local/bin
helm installed into /usr/local/bin/helm
version.BuildInfo{Version:"v3.21.2", GitCommit:"125963406833fe0525be91f46c8b5b0f22fb9e32", GitTreeState:"clean", GoVersion:"go1.26.4"}
root@ubuntu-2:~#
```

9.3. Устанавливаем Kompose

```
kompose version
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current				
			Dload	Upload	Total	Spent	Left	Speed			
0	0	0	0	0	--:--:--	--:--:--	--:--:--	0			
100	21.0M	100	21.0M	0	0	9330k	0	0:00:02	0:00:02	--:--:--	16.6M

```
1.34.0 (cbf2835db)
root@ubuntu-2:~#
```

10. Создаём проект prometheus-grafana

10.1. Создаём compose.yaml из прошлой работы

10.2. Создаём datasource для Grafana

10.3. Создаём prometheus.yml

Prometheus будет собирать метрики с Flask dev/prod на BM 10.0.0.100.

11. Генерируем Helm chart через Kompose

```
kompose convert --chart -f compose.yaml
ls -la
WARN Restart policy 'unless-stopped' in service grafana is not supported, convert it to 'always'
WARN Restart policy 'unless-stopped' in service blackbox is not supported, convert it to 'always'
WARN Restart policy 'unless-stopped' in service prometheus is not supported, convert it to 'always'
WARN File don't exist or failed to check if the directory is empty: stat :/var/lib/grafana: no such file or directory
WARN File don't exist or failed to check if the directory is empty: stat :/prometheus: no such file or directory
INFO Kubernetes file "compose/templates/blackbox-service.yaml" created
INFO Kubernetes file "compose/templates/grafana-service.yaml" created
INFO Kubernetes file "compose/templates/prometheus-service.yaml" created
INFO Kubernetes file "compose/templates/blackbox-deployment.yaml" created
INFO Kubernetes file "compose/templates/grafana-deployment.yaml" created
INFO Kubernetes file "compose/templates/grafana-data-persistentvolumeclaim.yaml" created
INFO Kubernetes file "compose/templates/grafana-cm0-configmap.yaml" created
INFO Kubernetes file "compose/templates/prometheus-deployment.yaml" created
INFO Kubernetes file "compose/templates/prom-data-persistentvolumeclaim.yaml" created
INFO Kubernetes file "compose/templates/prometheus-cm0-configmap.yaml" created
INFO chart created in "compose/"
total 24
drwxr-xr-x 5 root root 4096 июл  3 10:23 .
drwxr-xr-x 3 root root 4096 июл  3 10:22 ..
drwxr-xr-x 3 root root 4096 июл  3 10:23 compose
-rw-r--r-- 1 root root  793 июл  3 10:22 compose.yaml
drwxr-xr-x 2 root root 4096 июл  3 10:22 grafana
drwxr-xr-x 2 root root 4096 июл  3 10:22 prometheus
root@ubuntu-2:~/lab8/prometheus_grafana#
```

```
root@ubuntu-2:~/lab8/prometheus_grafana# mv compose promgra
root@ubuntu-2:~/lab8/prometheus_grafana# ls -la
total 24
drwxr-xr-x 5 root root 4096 июл  3 10:24 .
drwxr-xr-x 3 root root 4096 июл  3 10:22 ..
-rw-r--r-- 1 root root  793 июл  3 10:22 compose.yaml
drwxr-xr-x 2 root root 4096 июл  3 10:22 grafana
drwxr-xr-x 2 root root 4096 июл  3 10:22 prometheus
drwxr-xr-x 3 root root 4096 июл  3 10:23 promgra
root@ubuntu-2:~/lab8/prometheus_grafana#
```

12. Дорабатываем Helm chart

12.1. Chart.yaml

12.2. values.yaml

12.3. ConfigMap для Grafana

12.4. ConfigMap для Prometheus

12.5. Grafana Deployment

12.6. Prometheus Deployment

12.7. Blackbox Deployment

12.8. Grafana Service с LoadBalancer и ExternalIP

12.9. Prometheus Service

12.10. Blackbox Service

13. Проверяем Helm chart

0 chart(s) failed:

```
root@ubuntu-2:~/lab8/prometheus_grafana# helm lint ./promgra
==> Linting ./promgra
[INFO] Chart.yaml: icon is recommended

1 chart(s) linted, 0 chart(s) failed
root@ubuntu-2:~/lab8/prometheus_grafana#
```

Подстановка IP портов:

helm template promgra ./promgra

```
1 chart(s) linted, 0 chart(s) failed
root@ubuntu-2:~/lab8/prometheus_grafana# helm template promgra ./promgra
---
# Source: promgra/templates/grafana-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: grafana-cm0
  labels:
    io.kompose.service: grafana
data:
  datasource.yml: |
    apiVersion: 1
    datasources:
      - name: Prometheus
        type: prometheus
        url: http://prometheus:9090
        isDefault: true
        access: proxy
        editable: true
---
# Source: promgra/templates/prometheus-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: prometheus-cm0
  labels:
    io.kompose.service: prometheus
data:
  prometheus.yml: |
    global:
      scrape_interval: 5s
      scrape_timeout: 3s
      evaluation_interval: 15s

    alerting:
      alertmanagers:
        - static_configs:
            - targets: []

    rule_files: []

    scrape_configs:
      - job_name: prometheus
        static_configs:
          - targets: ['localhost:9090']

      - job_name: blackbox-http
        metrics_path: /probe
        params:
          module: [http_2xx]
        static_configs:
          - targets:
              - http://10.0.0.100:54321
              - http://10.0.0.100:54000
              - https://student.psu.ru
        relabel_configs:
          - source_labels: [__address__]
            target_label: __param_target
          - source_labels: [__param_target]
            target_label: instance
          - target_label: __address__
            replacement: blackbox:9115

      - job_name: view_total
        metrics_path: /metrics
        scrape_interval: 15s
```


14. Упаковываем chart

```
root@ubuntu-2:~/lab8/prometheus_grafana# cd ~/lab8/prometheus_grafana

helm package ./promgra

ls -la *.tgz
Successfully packaged chart and saved it to: /root/lab8/prometheus_grafana/promgra-0.1.0.tgz
-rw-r--r-- 1 root root 1739 июл 3 10:30 promgra-0.1.0.tgz
root@ubuntu-2:~/lab8/prometheus_grafana#
```

15. Запускаем tunnel на VM 10.0.0.110

```
user@ubuntu-2:~$ sudo su
root@ubuntu-2:/home/user# sudo pkill -f "minikube tunnel" || true

sudo nohup minikube tunnel --bind-address=10.0.0.110 > /tmp/minikube-tunnel.log 2>&1 &

sleep 5

tail -n 20 /tmp/minikube-tunnel.log
[1] 19941
nohup: ignoring input
* Tunnel successfully started

* NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...

root@ubuntu-2:/home/user#
```

16. Устанавливаем Helm release

```
helm upgrade --install promgra ./promgra-0.1.0.tgz \
  -n monitoring \
  --create-namespace

kubectl get all -n monitoring
kubectl get services -n monitoring
kubectl get pods -n monitoring -o wide
Release "promgra" does not exist. Installing it now.
NAME: promgra
LAST DEPLOYED: Fri Jul 3 10:32:52 2026
NAMESPACE: monitoring
STATUS: deployed
REVISION: 1
TEST SUITE: None

NAME                READY   STATUS              RESTARTS   AGE
pod/blackbox-58ccc48897-bbz7q   0/1     ContainerCreating   0           0s
pod/grafana-55bbfdd4d6-qgrnw    0/1     ContainerCreating   0           0s
pod/prometheus-84d5bb5dbd-v9jj5 0/1     ContainerCreating   0           0s

NAME                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/blackbox    ClusterIP           10.105.45.158   <none>           9115/TCP         0s
service/grafana     LoadBalancer       10.96.26.228    10.0.0.110       3000:30689/TCP   0s
service/prometheus  ClusterIP           10.100.46.39    <none>           9090/TCP         0s

NAME                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/blackbox  0/1     1             0           0s
deployment.apps/grafana  0/1     1             0           0s
deployment.apps/prometheus 0/1     1             0           0s

NAME                DESIRED   CURRENT   READY   AGE
replicaset.apps/blackbox-58ccc48897 1         1         0       0s
replicaset.apps/grafana-55bbfdd4d6   1         1         0       0s
replicaset.apps/prometheus-84d5bb5dbd 1         1         0       0s

NAME                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
blackbox            ClusterIP           10.105.45.158   <none>           9115/TCP         0s
grafana             LoadBalancer       10.96.26.228    127.0.0.1,10.0.0.110 3000:30689/TCP   0s
prometheus          ClusterIP           10.100.46.39    <none>           9090/TCP         0s

NAME                READY   STATUS              RESTARTS   AGE   IP              NODE     NOMINATED NODE   READINESS GATES
blackbox-58ccc48897-bbz7q 0/1     ContainerCreating   0           0s    <none>          minikube <none>          <none>
grafana-55bbfdd4d6-qgrnw  0/1     ContainerCreating   0           0s    <none>          minikube <none>          <none>
prometheus-84d5bb5dbd-v9jj5 0/1     ContainerCreating   0           0s    <none>          minikube <none>          <none>
```

Контейнеры запустились:

```
Date: Fri, 03 Jul 2026 10:34:20 GMT

root@ubuntu-2:~/lab8/prometheus_grafana# kubectl get pods -n monitoring -o wide
kubectl get services -n monitoring

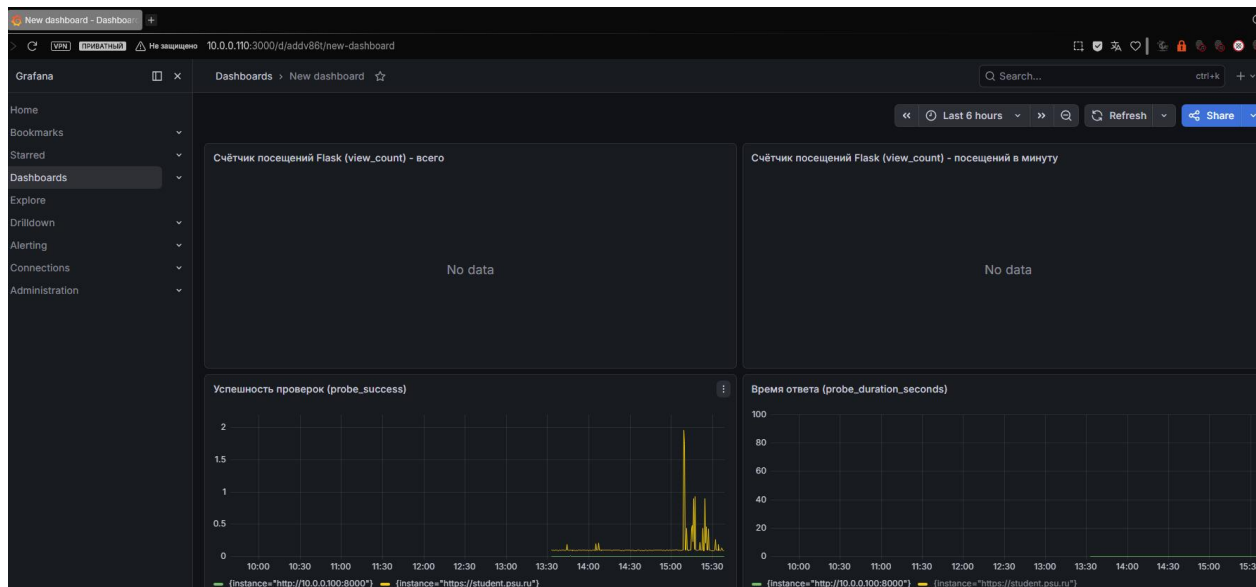
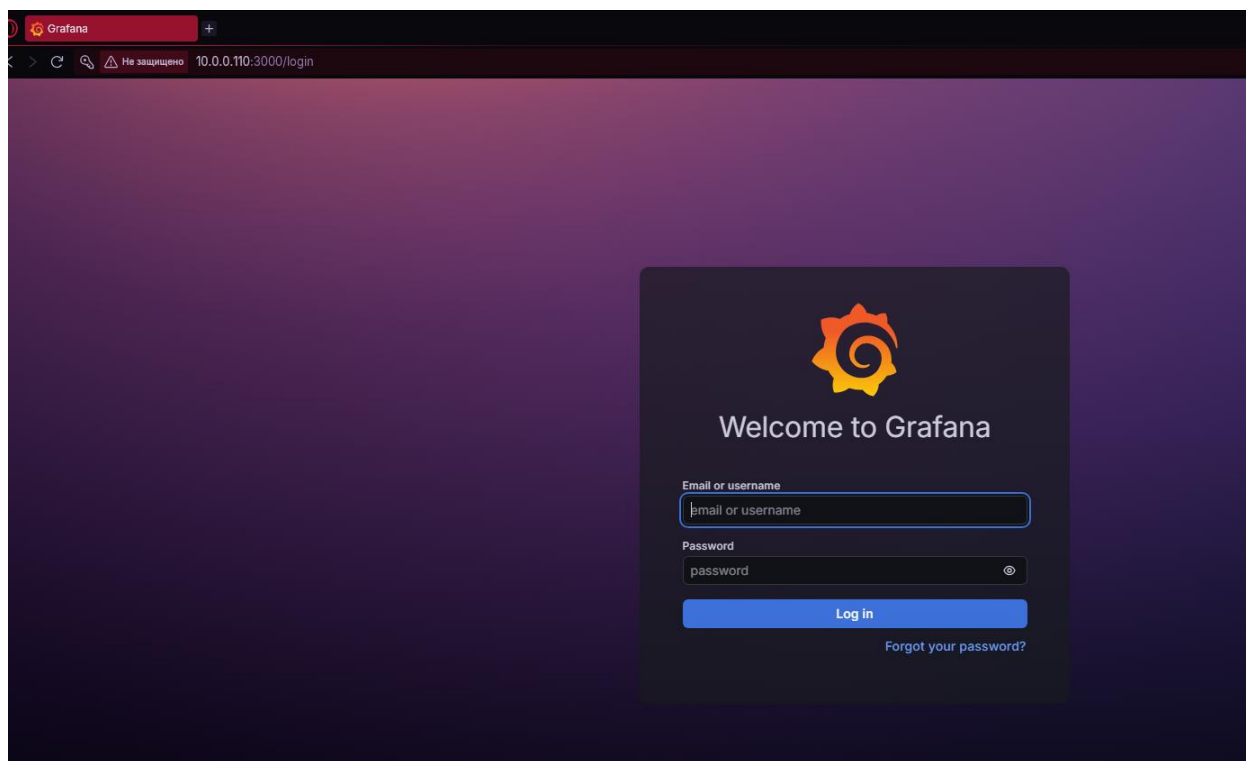
NAME                READY   STATUS    RESTARTS   AGE   IP              NODE     NOMINATED NODE   READINESS GATES
blackbox-58ccc48897-bbz7q 1/1     Running   0           2m50s  10.244.0.5      minikube <none>          <none>
grafana-55bbfdd4d6-qgrnw  1/1     Running   0           2m50s  10.244.0.4      minikube <none>          <none>
prometheus-84d5bb5dbd-v9jj5 1/1     Running   0           2m50s  10.244.0.3      minikube <none>          <none>

NAME                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
blackbox            ClusterIP           10.105.45.158   <none>           9115/TCP         2m50s
grafana             LoadBalancer       10.96.26.228    127.0.0.1,10.0.0.110 3000:30689/TCP   2m50s
prometheus          ClusterIP           10.100.46.39    <none>           9090/TCP         2m50s
root@ubuntu-2:~/lab8/prometheus_grafana#
```

17. Проверяем Grafana

curl -I <http://10.0.0.110:3000/login>

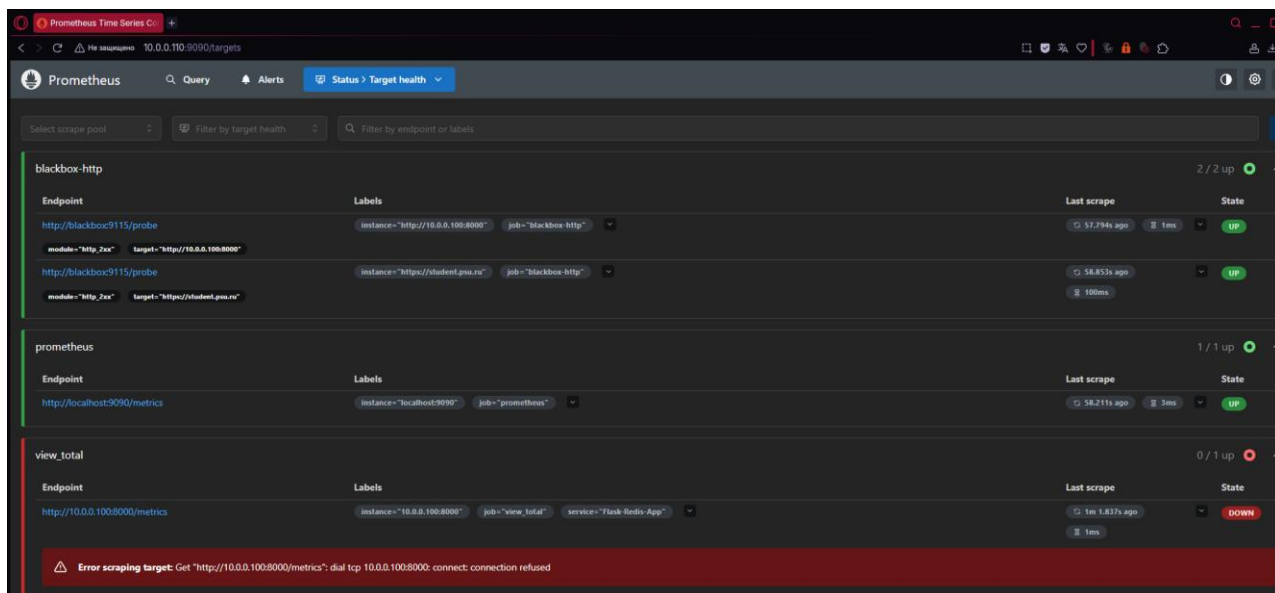
```
root@ubuntu-2:~/lab8/prometheus_grafana# curl -I http://10.0.0.110:3000/login
HTTP/1.1 200 OK
Cache-Control: no-store
Content-Type: text/html; charset=UTF-8
X-Content-Type-Options: nosniff
X-Frame-Options: deny
X-Xss-Protection: 1; mode=block
Date: Fri, 03 Jul 2026 10:37:07 GMT
```



18. Проверяем Prometheus

Чтобы открыть Prometheus UI, можно временно пробросить порты

```
kubectl -n monitoring port-forward --address 0.0.0.0 svc/prometheus 9090:9090
```



Flask dev/prod в состоянии UP

19. Проверка Helm values

```
helm get values promgra -n monitoring --all
USER-SUPPLIED VALUES:
null
COMPUTED VALUES:
EXTERNAL_IP: 10.0.0.110
EXTERNAL_PORT: 3000
FLASK_DEV_TARGET: 10.0.0.100:54321
FLASK_PROD_TARGET: 10.0.0.100:54000
GF_ADMIN_PASSWORD: HiGrafana
root@ubuntu-2:~/lab8/prometheus_grafana#
```